

Workflow



The central logic of the project:

understand the current state (EDA)

→ **discover structural differences (clustering)**

→ **identify what drives growth (modelling)**

→ **point to concrete actions (scenario simulation)**

every phase feeds directly into the final policy recommendations.

Phase 0: Data Preparation

Objective: Map all 83 columns to DAG nodes and produce a clean analytical dataset

Task	Detail
DAG node mapping	Assign each column to a node (Housing / Education / Transport, etc.)
Missing value treatment	GVA missing for 105 LSOAs, VOA for 83 — decide on imputation or exclusion strategy
Feature engineering	Combine columns within the same node (e.g. aggregate Level 1–4 qualification rates into a single skills score)
Normalisation	Align scales (per-capita, percentage, Z-score)
Geospatial join	Link the dataset to LSOA shapefiles in preparation for mapping

Phase 1: EDA

Objective: Understand distributions, spatial patterns, and pinpoint Swindon's specific weaknesses

- **Univariate analysis:** Distribution plots for each DAG node variable
- **Spatial visualisation:** Choropleth maps of GVA, IMD, skills, and enterprise density — identifying the weakest LSOAs within Swindon
- **Correlation matrix:** Cross-node correlations to validate DAG assumptions
- **Swindon vs benchmark:** Systematic comparison of Swindon's 140 LSOAs against the 989 surrounding LSOAs (Berkshire / Oxfordshire)

- **Gap identification:** Is Swindon's primary weakness in skills, enterprise density, or transport connectivity?

Phase 2: Place Typology

Objective: Segment all 1,129 LSOAs into distinct economic types rather than treating them uniformly

- **Method:** K-Means or Hierarchical Clustering using upstream DAG node variables as inputs
- **Output:** A profile for each cluster (e.g. high-skills / high-GVA; low-skills / high-deprivation; industrial employment base)
- **Key question:** Which cluster types do Swindon's 140 LSOAs fall into, and in what proportions?
- **Policy value:** Different place types require different intervention strategies

Phase 3: Modelling

Objective: Use interpretable machine learning to quantify each DAG node's contribution to GVA

- **Target variables:** GVA (primary), employment rate, IMD (secondary)
- **Input features:** Upstream DAG nodes (education, digital connectivity, business structure, transport, etc.)
- **Model choice:** XGBoost / Random Forest for predictive power + SHAP for interpretability
- **Analysis:**
 - Global SHAP importance: which factor has the greatest impact on GVA across all LSOAs?
 - Local SHAP analysis: which variables are dragging down GVA in Swindon's weakest LSOAs?
 - Cluster-stratified modelling: do growth drivers differ by place type?
- **Caution:** Mediator variables (Employment Rate, Productivity) must be handled in separate model layers to avoid confounding

Phase 4: Sector and SME Analysis

Objective: Identify which industries and enterprise types are Swindon's growth engines

- **Sector spatial analysis:** Geographic distribution of Creative, KIBS, Industrial, and Green-aligned enterprises across Swindon's LSOAs
- **Sector–GVA relationship:** Which sector concentrations are associated with higher GVA?
- **SME growth modelling:** Use VOA rateable value change (2023 → 2026) as a proxy for SME activity; identify LSOAs with the highest growth potential
- **Business size structure:** How does the micro / small / medium / large enterprise mix affect local economic resilience?

Phase 5: Scenario Simulation

Objective: Provide the council with evidence on "if we do X, how much could GVA increase?"

- **Method:** Counterfactual analysis using the Phase 3 model
- **Example scenarios:**
 - If Swindon's Level 4+ qualification rate rises from 21% to the regional mean of 32%, what is the projected GVA change?
 - If enterprise density increases from 161 to 200 per 1,000 residents, how does employment rate respond?
 - If broadband coverage improves in poorly connected LSOAs, what is the estimated GVA uplift?
- **Output:** Prioritised intervention list with estimated GVA impact per lever

Phase 6: Reporting

Output	Audience	Content
Technical Notebook	Data science team	Full code, methodology, model evaluation
Policy Summary (2–4 pages)	Council decision-makers	Key findings and recommendations in plain language, supported by maps
Interactive Dashboard (optional)	Both audiences	Interactive choropleth maps, cluster profiles